

18 $2x^2 + 12x - 2$ can be written in the form $a(x+b)^2 - c$.

Find the values of a , b and c .

$$= \frac{b}{2}$$

$$: 6^2 = 36$$

$$2x^2 + 12x + 36 - 38 - 36$$

$$= \frac{22}{2}$$

$$2(x^2 + 6x + 18) - 38$$

$$= 6$$

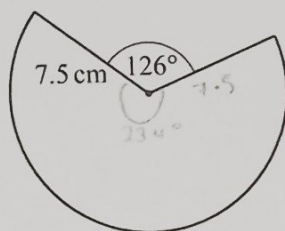
$$2[(x+6)^2] - 38$$

$$= 2x^2 + 24x + 6 - 38$$

$$=$$

$$a = \dots\dots\dots 2 \dots\dots\dots, b = \dots\dots\dots 6 \dots\dots\dots, c = \dots\dots\dots 38 \dots\dots\dots [3]$$

19



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The diagram shows a major sector of a circle with radius 7.5 cm.

Calculate the perimeter of the major sector.

$$= \frac{360}{360} \times 2\pi r$$

$$= \frac{234}{360} \times 2\pi \times 7.5$$

$$= 30.6 \text{ cm}$$

$$= 30.6 + 7.5 + 7.5$$

$$= 45.6 \text{ cm}$$

$$\dots\dots\dots 45.6 \dots\dots\dots \text{ cm } [4]$$